

Stopping Pump Failures Before They Happen

Business case for the CFO and Board | Kenya Pipeline Company (KPC) | Team NULL_TERMINATORS, Week 11

THE ASK: Approve a **KES 5 million, 3-month pilot at one station**. If it works, roll out to all stations for a further KES 10 million. **Full-scale payback: about 7 months.**

1. The Problem

KPC's pumps push fuel through 1,342 km of pipeline, yet they are serviced on a fixed calendar, not according to their condition. Healthy pumps are serviced needlessly, while worn ones can fail between visits. A failure halts deliveries, forces costly emergency repairs and can lead to leaks. The stakes are large: the 2018 Kiboko spill lost 551,000 litres of fuel, worth about **KES 63 million** in product alone. A hairline leak left unnoticed for a single day loses roughly **KES 14 million** of fuel. The 2011 Sinai fire began with a faulty part worth under KES 1,000; it cost KPC over **KES 103 million** directly, and about 100 people died.

2. The Proposed Solution

Fit low-cost sensors to each pump to track vibration, temperature, pressure and electrical load. The system learns what a healthy pump looks like, then every day (a) rates each pump's chance of failing in the next 7 days, (b) estimates how many hours of life it has left, and (c) gives planners one ranked repair list on a single screen, with email alerts. Weather and local activity around each station are factored in. In a trial on simulated data for 24 pumps, it **caught 84% of failures with only 2.4% false alarms** (about 1 in 40 alerts). Confirming this on KPC's real pumps is the purpose of the pilot.

3. Financial Impact (annual, KES millions, with the math)

Where the savings come from	Calculation	KES M / yr
Fewer emergency repairs and delivery stoppages	6 failures/yr x 84% caught early x (KES 4.0M emergency cost - KES 1.5M planned cost)	12.6
Fuel loss avoided	One Kiboko-size loss (KES 63M) every 3 years = KES 21M/yr x 50% prevented	10.5
Less unnecessary servicing	60 services/yr x 25% skipped x KES 0.4M each	6.0
Total annual benefit	12.6 + 10.5 + 6.0	29.1
Less: running costs (cloud, sensor upkeep, support)		-4.0
Annual savings (net)	29.1 - 4.0	25.1

KES 15M

One-off investment: KES 5M pilot + KES 10M rollout (30 pumps x KES 0.25M sensors = 7.5M; setup and training = 7.5M)

~7 months

Payback period
15.0 / 25.1 = 0.6 years

223%

3-year return on investment
(87.3 - 27.0) / 27.0
benefits 3 x 29.1; costs 15.0 + 3 x 4.0

Stress test: if benefits are only half of the above, payback is still about 17 months and the 3-year return is 62%. **Not counted:** clean-up costs, legal claims and lives protected, which could each exceed the figures above. **Basis:** fuel value uses KPC's documented Kiboko incident and EPRA's 2026 fuel costs; the fleet size (30 pumps), failure count and repair costs are planning assumptions to be replaced with KPC actuals early in the pilot.

4. Top Two Risks and How We Handle Them

Risk 1: Results not yet proven on KPC's real pumps. Our 84% figure comes from simulated data, so real-world accuracy could be lower.

Our response: test at one station first, so only KES 5M is at stake. The remaining KES 10M is released only if the pilot catches at least 80% of failures with no more than 5% false alarms.

Risk 2: Planners may not trust or act on the alerts, or data access may be slow.

Our response: every alert says in plain words why a pump is flagged and how many hours it has left; we run alongside the current schedule during the pilot; and we agree a KPC engineering contact and data access before starting.

RECOMMENDATION: GO (pilot first, then decide)

Approve **KES 5 million and 3 months** for a one-station pilot, and nominate a KPC engineering contact with access to historical pump failure records. At the end of the pilot we return to this Board with real results, and the KES 10 million fleet-wide rollout goes ahead only if the targets in Risk 1 are met.

Sources: Breaking Kenya News (2020); Pipeline and Gas Journal (2019); Nation (2020); EPRA (2026); team prototype results (Week 6).